



# Making predictions

In Bayesian inference, we maintain a full distribution over parameters, reflecting our uncertainty. We make a prediction for a new observation  $y^*$ , then we average all possible outcomes weighted by how likely each  $\theta$  is.

## Posterior predictive distribution

$$P(y^*|y) = \int P(y^*|\theta) P(\theta|y) d\theta$$

- we observed  $N=123$  views  $\wedge y=12$  clicks
- new campaign  $N^*=50$  views
- ? we want to know  $P(y^*|y=12)$  i.e. the probability distribution over future clicks.
- we assume: click follow a Bernoulli trial w/ unknown click-through rate  $\theta$ .
- $\theta \sim \text{Beta}(a, b)$  is our prior

## a) Joint distribution / model

- we assume data  $y$ , future observations  $y^*$  and latent parameter  $\theta$ , and start w/ joint distribution over all the parameters

$$P(y^*, y, \theta) = P(y^*|\theta) \overset{P(y|N, \theta)}{=} P(y|\theta) \overset{P(\theta|a, b)}{=} P(\theta)$$

b) condition over the data - we are asking: given the data we have seen ( $y=12$ ), what should I believe about unknown parameters or future outcomes?

$$P(y^*, \theta|y) = \frac{P(y^*, y, \theta)}{P(y)} = \frac{P(y^*|\theta) \cdot P(y|\theta) \cdot P(\theta)}{P(y)}$$

$P(y) = \int P(y|\theta) P(\theta) d\theta$

→ we want only  $P(y^*|y)$  - which describes given all observed data, what should I believe about new observations. This gives us the distribution over future data, based on all evidence and all possible plausible  $\theta$  values.



We do not remove  $\theta$  because it is important, we do so bcs we've already used the data to update our beliefs about it. Now we just want to pay attention to what actually matters  $\Rightarrow$  new data  $y^*$

## c) marginalization and obtaining posterior predictive distribution

$$P(y^*|y) = \int P(y^*, \theta|y) d\theta = \int P(y^*|\theta) P(\theta|y) d\theta$$

- this integral can be calculated as expectation  $P(y^*|y) = \mathbb{E}_{\theta \sim P(\theta|y)} [P(y^*|\theta)]$  - where we average the likelihood for the new point under the posterior over  $\theta$ .
- naïve distribution supplies the weights (here it is the posterior  $P(\theta|y)$ )
- 1. we draw  $\theta$  from  $P(\theta|y)$
- 2. draw/evaluate a future observation from  $P(y^*|\theta)$
- 3. repeat (1) & (2) many times: the empirical frequencies you get for  $y^*$  converge to  $P(y^*|y)$

# Setting up the model

- likelihood (Bernoulli trials)

$$p(y|\theta) = \text{Bin}(y|N, \theta) = \binom{N}{y} \theta^y (1-\theta)^{N-y}$$

- prior (conjugate Beta)

choose uniform Beta(1,1)  $p(\theta) = \text{Beta}(\theta|a_0=1, b_0=1)$

- posterior over  $\theta$

$$p(\theta|y) = \text{Beta}(\theta|a_0+y, b_0+N-y) = \text{Beta}(\theta|13, 112)$$

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## d) Predictive for one new Bernoulli trial

- we start from the posterior predictive

$$p(y^*=1|y) = \int_0^1 \underbrace{p(y^*=1|\theta)}_{\text{likelihood}} \underbrace{p(\theta|y)}_{\text{posterior}} d\theta$$

- now we identify each part of the integrand

$$p(y^*=1|\theta) \Rightarrow \theta$$

$$p(\theta|y) \Rightarrow \text{Beta}(\theta|a,b) \text{ where } a=13, b=112$$

Plugging these in

$$p(y^*=1|y) = \int_0^1 \theta \text{Beta}(\theta|a,b) d\theta$$

## e) Predictive for 50 new trials

— because of cond. indep.,  $y^* | \theta \sim \text{Bin}(N^*, \theta)$ , after marginalizing w.r.t. posterior we obtain Beta Binomial dist.

$$p(y^* | y) = \text{Beta Bin} \left( y^* | N^* = 50, a=13, b=12 \right) = \binom{50}{y^*} \frac{B(13+y^*, 12+50-y^*)}{B(13, 12)}$$

• pred mean  $E[y^* | y] = N^* \frac{13}{13+12} = 50 \cdot 0.103 \approx \underline{\underline{5.15}}$

a) posterior for  $\theta$   $p(\theta | y) = \mathcal{N}(\bar{y}, \frac{\sigma^2}{n})$

b) exact posterior - predictive distribution, we need  $p(y^* | y) = \int p(y^* | \theta) p(\theta | y) d\theta$   
 $\Rightarrow y^* | y \sim \mathcal{N}(\bar{y}, \sigma^2 + \frac{\sigma^2}{n})$   
 → variance gets extra  $\frac{\sigma^2}{n}$  to reflect uncertainty in  $\theta$ .  
 ↑  $p(y^* | \theta) = \mathcal{N}(y^*, \sigma^2)$

c) plug-in (Dirac  $\delta$ ) predictive

approximate the posterior with  $p(\theta | y) \approx \delta(\theta - \bar{y})$  with  $\theta = \bar{y}$

Sifting property collapses the integral:

$$p(y^* | y) \approx p(y^* | \theta) = \mathcal{N}(\bar{y}, \sigma^2)$$

no extra variance term, we ignore uncertainty in  $\theta$

Task 1.1 use a product rule to write down the joint distribution of  $y, a, b$ , given  $X$

$$p(y, a, b | x) = p(y | a, b, x) p(a, b)$$

Task 1.2

$$\begin{aligned} \sum_{y^*} p(y, y^*, a, b | x, x^*) &= p(y, a, b | x) \sum_{y^*} p(y^* | y) \\ &= p(y, a, b | x) \end{aligned}$$

Task 1.3 Use Bayes rule to express the posterior  $p(\alpha, \beta | y, x)$  in terms of the prior, likelihood and evidence and argue that the posterior is  $\propto$  to the joint  $p(y, \alpha, \beta, x)$

- use product rule:

$$p(y, \alpha, \beta | x) = p(y | \alpha, \beta, x) \cdot p(\alpha, \beta) \Rightarrow \text{we plug this into the Bayes Rule.}$$

$$p(\alpha, \beta | y, x) = \frac{p(y, \alpha, \beta | x)}{p(y | x)} = \frac{p(y | \alpha, \beta, x) \cdot p(\alpha, \beta)}{p(y | x)}$$

- because the evidence  $p(y | x)$  is  $\perp$  of the parameters, the post. is prop to joint lik.

$$p(\alpha, \beta | y, x) \propto p(y, \alpha, \beta | x) = p(y | \alpha, \beta, x) \cdot p(\alpha, \beta)$$

# Dataset

- we have a historical data from 23 earlier shuttle launches

→ each provides

- $t_i$  - the temperature at the time of launch
- $y_i$  - the # of failed O-rings on that launch (0-6)
- All rockets had  $N=6$  O-rings, so the possible values of  $y_i$  are  $\{0, \dots, 6\}$

∴ our dataset  $\mathcal{D} = \{(t_i, y_i)\}_{i=1}^M$ ,  $M=23$

- we have 23 rows in total

→ each row is a pair:

- $t_i$ : temperature during launch
- $y_i$ : # of O-rings that failed during that launch

We assume:

- the probability of an O-ring failure depends on temperature
- lower temp increases a chance of failure

- we are modelling the # of O-ring failures (successes in our case) observed at given temperature  $x_i$ . Hence the # of failed O-rings  $y_i$  for a given temp  $x_i$  can be described by Binomial Distribution, w/  $N_i=6$  and prob  $\theta_i \in [0,1]$ .

$$y_i | \theta_i \sim \text{Bin}(N_i, \theta_i)$$

PMF for Binomial model

$$P(y_i | \theta_i) = \binom{6}{y_i} \theta_i^{y_i} (1-\theta_i)^{6-y_i}$$

# of failed O-rings (points to  $y_i$ )  
total # of O-rings on the rocket (Vn6) (points to 6)  
Prob of failure for a single O-ring (we assume it's the same for each one at a given temp) (points to  $\theta_i$ )

the following likelihood tells us how likely it is to see  $y_i$  failures if the failure rate is  $\theta$ .

example: on day  $i$ , the temperature was  $x_i = 31^\circ\text{F}$ , and we observed  $y_i = 2$  failed O-rings. Assume the failure prob at this temperature is  $\theta = 0.3$ , then:  $P(y_i=2 | \theta_i=0.3) = \binom{6}{2} (0.3)^2 (0.3)^4 = \underline{0.32}$

# Logistic Link

$$\theta_i = \sigma(\alpha + \beta x_i)$$

intercept  
converts the results to [0,1] interval     slope

- we want to  $\theta_i$  vary smoothly w/ temperature, but never leave  $[0,1]$ . Mapping a linear predictor  $\alpha + \beta x_i$  through the sigmoid  $\sigma$  does exactly that

example:

$$\alpha = -7, \beta = 0.25, x_i = 30$$

$$z = -7 + 0.25 \cdot 30 = 0.5 \Rightarrow \theta_i = \sigma(0.5) \approx 0.622$$

## Prior distributions

$$\alpha \sim \mathcal{N}(0, \sigma_\alpha^2) \quad \beta \sim \mathcal{N}(0, \sigma_\beta^2)$$

- we believe, before seeing the data, that each coefficient is near zero with uncertainty  $\sigma_\alpha, \sigma_\beta$
- we expect  $\alpha$  &  $\beta$  to be near 0 before seeing the data
- prevents model from learning extreme parameters w/ limited data
- $p(\alpha, \beta | \text{data}) \propto p(\text{data} | \alpha, \beta) \cdot p(\alpha, \beta)$

## Data likelihood

$$L(\theta) = p(\text{data} | \theta)$$

all unknown quantities ( $\alpha, \beta$ ) in our case  
everything observed

our data density for a single obs

$$p(y_i | x_i, \alpha, \beta) = \text{Bin}(y_i | N, \sigma(\alpha + \beta x_i))$$

product rule, if  $A$  &  $B$  are independent, given  $C$ , then

$$p(A, B | C) = p(A | C) p(B | C), \text{ applying this to all data pts, we get:}$$

$$p(y_1, \dots, y_n | x, \alpha, \beta) = \prod_{i=1}^n p(y_i | x_i, \alpha, \beta) \Rightarrow \text{full data likelihood}$$

$$\Rightarrow p(y | x, \alpha, \beta) = \prod_{i=1}^n \binom{N}{y_i} \left[ \sigma(\alpha + \beta x_i) \right]^{y_i} \left[ 1 - \sigma(\alpha + \beta x_i) \right]^{N - y_i}$$

- usually during numerical calculation, we calculate in log space

$$\ell(\alpha, \beta) = \log p(y | x, \alpha, \beta) = \sum_{i=1}^n \left[ \log \binom{N}{y_i} + y_i \log \sigma(f_i) + (N - y_i) \log [1 - \sigma(f_i)] \right]$$

f\_i = \alpha + \beta x\_i



